

↳ Previously we described languages using phrases to describe a language.

1. $L = \{x^n \text{ for } n = 1, 2, 3, \dots\}$

↳ Means all combination of x except Λ .

2. $L = \{x^n \text{ for } n = 3, 4, 8, 22, \dots\}$

↳ No clear pattern. We are simply guessing the next numbers. Therefore, we need to impose mathematical definition to avoid this.

- The term "imprecise" and "guesswork" refers to the general approach of defining a language by listing specific elements (like $\{3, 4, 8, 22, \dots\}$) without providing a clear, general rule or pattern. In automata theory, we aim to define languages in a precise way so that we can create automata to recognize them.
- When you just list a few elements (without a pattern), it's like guessing the next number in the sequence. Without a formal rule, it's unclear what the rest of the language should look like, and it's impossible to predict whether you've captured all the valid strings. This kind of definition is imprecise because:
 - No rule or pattern: Without a general formula or pattern, you can't reliably define the language.
 - Can't automate: The lack of a clear pattern makes it impossible to create a finite automaton or other formal machine to recognize all valid strings of the language.
- So, it's called "guesswork" because you are not using a rigorous mathematical definition—you're simply listing numbers you think belong to the language, but without a clear underlying principle that guarantees completeness.

↳ Let's use a new way to define a language which is more precise.

↳ We will define Kleene plus closure over letters instead of on a set.

Example,

$\Sigma = \{a, b\}$

↳ $L = \{a, ab, abb, abbb, \dots\}$

Regular Expression: ab^*

Descriptive definition: All words that have at least one a & any number of b including none.

↳ a^*b^* means a must come before b.

Plus operator:

$$\hookrightarrow E = x + y$$

↳ choose either x or y but not both

Regular Expression:

↳ Every letter from Σ can be made into a R.E

↳ Λ is a R.E

Union of two languages:

$$L_1 = (a+b)^* a (a+b)^*$$

$$L_2 = b^*$$

$$L_1 \cup L_2 = (a+b)^* a (a+b)^* + b^*$$

Example,

$$L_1 = \{a, aa, aaa\}, L_2 = \{b, bb, bbb\}$$

$$L_1 + L_2 = \{a, aa, aaa, b, bb, bbb\} \rightarrow \text{combined into one large set. Now, choose one word.}$$

Important:

↳ Expression can be distributed but operation cannot be

$$(a+c)b^* = ab^* + cb^*$$

$$(ab)^* = a^*b^*$$

↳ Not commutative
 $ab \neq ba$

Product of two sets:

$$S = \{a, aa, aaa\}, T = \{bb, bbb\}$$

$$ST = \{abb, abbb, aabb, aabbb, aaabb, aaabbb\}$$

Chapter 04

Q2. $(aaa+b)^*$

Q4. $a^*ba^*ba^*ba^*$

Q5. $a^*ba^*ba^*(b+\wedge)a^*$

Q6. (i) $(a+b)^*(aa+bb)$

(ii)

Q7.

Q8.